

Multi-Vendor Interoperable 800G Optical Communications over 1602-km through 800G-OpenROADM, 800-ZR and 800GBASE-2xFR4 Fiber Links

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Abstract: For the first time, we report a multi-vendor interoperable optical transmission using 800G transceivers compliant with the 800G-OpenROADM, 800-ZR and 800GBASE specifications over SSMF line sections of 15x100-km, 100-km and 2-km, respectively. The OpenROADM-compliant probabilistic constellation shaping is used for the first time. © 2026 The Author(s)

1. Introduction

Interoperability in optical communications is gaining momentum with the recent introduction of 800G pluggable optics which address various needs such as metropolitan area networks (MAN), data-center interconnects (DCI) and Ethernet client networks. Driven by IP-over-WDM applications, the development of cloud and AI services, as well as the need for ever-increasing data rate and cost/power-efficient optical networks, interoperable 800G optical transceivers are essential for the development of telecommunications in the years to come. The continuous progress of CMOS microelectronics (with 3-nm manufacturing process), and silicon/indium-phosphide photonic integrated circuits (PICs) is a key enabler for these 800G optics, which require a high level of integration, minimal energy consumption, and cost efficiency. The interworking of optical interfaces has been achieved through the combined efforts of standardization organizations such as IEEE [1], OIF [2], ITU-T [3], OpenROADM (OR) [4] and OpenZR+ [5] initially at 400G, now at 800G and in the future at 1600G. IEEE through the 802.3 recommendation [1] addresses the Ethernet client needs. OIF focuses on DCI through its 800-ZR specification [2]. OR [4] specifies WDM interfaces for MANs equipped with reconfigurable optical add/drop multiplexers (ROADM). Recently, OR released its new 800G specification [4], which introduces, for the first time, an interoperable probabilistic constellation shaping (PCS) [6-7], aiming to maintain the same performance level at 800G as at 400G without altering the use cases for internet service providers.

In this paper, we report, for the first time, a multi-vendor interoperable optical transmission which uses the latest developed 800G interoperable coherent pluggable optics, including 800-ZR 118-Gbaud dual-polarization (DP) 16QAM interfaces and, for the first time, 800G-OR 131-Gbaud DP-PCS-16QAM transceivers. The 2x400-GbE client data streams are transported end-to-end across three standard single mode fiber (SSMF) links of 100-km (800-ZR), 15x100-km (800G-OR), and 2-km (800GBASE-2xFR4), corresponding to the main 800G specifications released by the standardization organizations. The 800G-OR, 800-ZR, and 800GBASE-2xFR4 optics originate from various vendors and incorporate PIC and digital signal processing (DSP) chips (for 800-ZR and 800G-OR) supplied by various manufacturers, ensuring full interoperability of the presented experimental set-up in a multi-vendor environment. Error-free transmission over 1602 km was achieved over 24 hours, while the robustness to ROADM cascade insertion was investigated over the 800G-OR section.

2. Experimental set-up

The experimental set-up is depicted in Fig.1a). An 800-GbE tester capable of hosting digital coherent optics (DCO) generates the 2x400-GbE client data flows, which are converted into the optical domain by a first 800-ZR QSFP-DD coherent transceiver from vendor A, embedding a CMOS DSP chip manufactured by the vendor B (V#B). OIF selected DP-16QAM and O-FEC as modulation format and forward error correction (FEC) code for its 800-ZR multi-source agreement (MSA) [2]. The O-FEC [2-5] has a BER threshold of $\sim 1.9 \times 10^{-2}$, $\sim 15.3\%$ overhead and a net coding gain of ~ 11.6 -dB at $\text{BER} = 10^{-15}$. The 800-ZR symbol rate is ~ 118.2 -Gbaud. The ~ 0 -dBm 800-ZR signal at 1550.12-nm is injected into a 100-km ITU-T G.652 SSMF span. At the receiver, to detect the 800G signal, a second 800-ZR QSFP-DD optic, provided by the vendor#1 (V#1) and plugged in an evaluation board (EVB) of the same vendor, hosts a CMOS DSP ASIC also built by the V#1. Thus, two DSP chips from distinct vendors (V#B and V#1) face each other. The power consumption of the two 800-ZR QSFP-DD optics does not exceed ~ 27 -Watts

After optical-to-electrical conversion, the EVB extracts the client data frames and forwards them to an 800G-OR coherent QSFP-DD transceiver, also manufactured by the V#1, which implements the OR-compliant interoperable PCS [4, 6-7]. This optic converts the data streams into the optical domain. The coherent QSFP-DD interface incorporates a CMOS DSP ASIC designed and manufactured by the V#1, in accordance with the 800G-OR specifications [4,7]. The modulation format and FEC chosen by the OpenROADM MSA for 800G are DP-PCS-16QAM and O-FEC, respectively. The symbol rate and PCS entropy are ~ 131.3 -Gbaud and ~ 3.6 -bits/symbol, respectively. PCS is introduced at 800G to preserve the use cases defined for 400G-OR optical transport networks. The ~ 2.7 -dBm 800G DP-PCS-16QAM signal at 1550.12-nm (193.4-THz) is coupled with fifty-two 100G DP-QPSK channels operating at 32-Gbaud, spanning from 1533.9 to 1557.4-nm on the 50-GHz ITU grid. A second 800G DP-PCS-16QAM wavelength at 1548.91-nm (193.55-THz), generated by another QSFP-DD interface plugged in the V#1 EVB, is inserted at 150-GHz from the

first 800G channel. As shown in Fig.1b), the 800G signals undergo root-raised cosine (RRC) Nyquist filtering with a roll-off factor of ~ 0.125 [4]. The spectrum of the 52 WDM channels at 100G and the two 800G channels, recorded at the transmitter (TX) output, is shown in Fig.1d) with the red trace. Note that the power of each 800G signal (occupying a 150-GHz spectral slot) is three-fold that of a single 100G channel (occupying a 50-GHz spectral slot) to maintain a constant power spectral density (PSD) across the WDM comb. An optional programmable optical filter can be inserted at the output of the 1550.12-nm interface to emulate the optical transfer function of a wavelength selective switch (WSS) cascade, as shown in the inset of Fig. 1a), after measurement with a 150-MHz high-resolution optical spectrum analyzer. In this configuration, the channel under test is amplified by an EDFA, filtered by the simulated ROADM and attenuated by a variable optical attenuator (VOA). A third 800G-OR channel, generated by a coherent OSFP transceiver from another vendor (V#3), is inserted at 1551.32-nm (193.25-THz) in place of three 100G channels, with the aim of creating symmetrical interferers around the channel under test (see Fig. 1c). The transmission line of the OR section consists of fifteen 100-km spans of ITU-T G.652 SSMF. The span losses (~ 20 -dB) are compensated by single-stage EDFA with ~ 4.5 -dB noise figure. The signal spectrum recovered after 1500-km is shown in Fig.1d) with the blue trace.

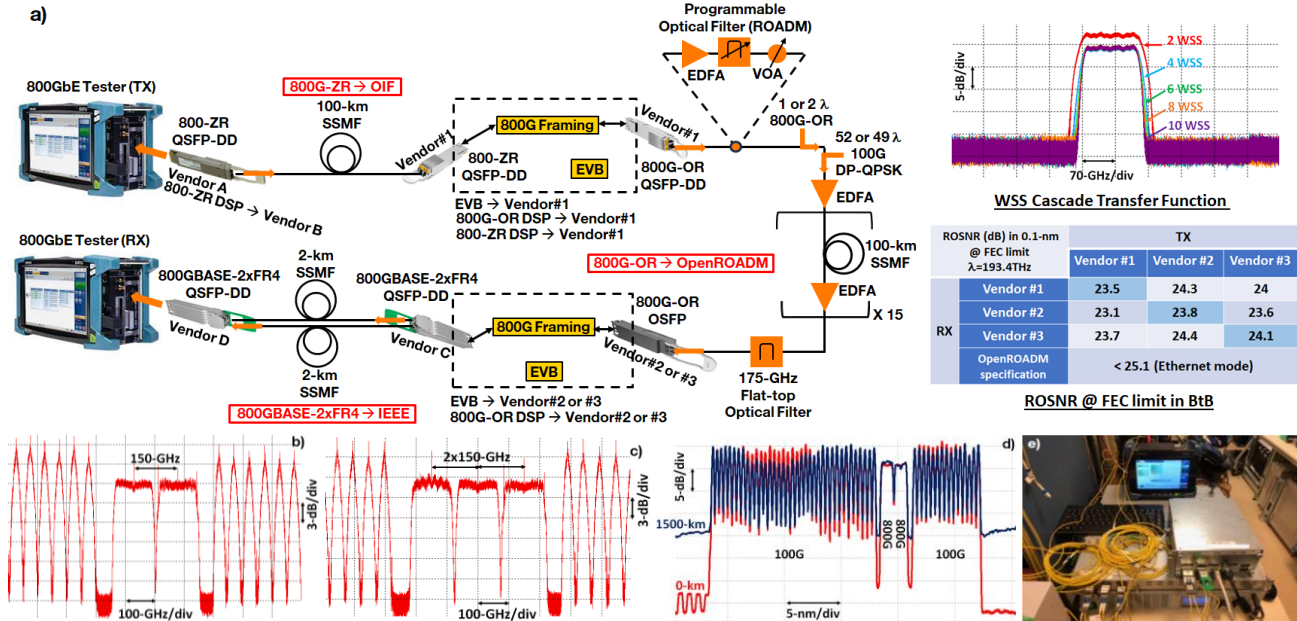


Fig.1: (a) Experimental set-up with the three interoperable fiber line sections (800-ZR, 800-OR, 800GBASE-2xFR4), with in insets the WSS cascade transfer function and the table giving the ROSNR @ FEC limit in BtB for the different configurations; (b) Zoom over the WDM comb where are located the two 150-GHz spaced 800-OR channels at the transmitter output; (c) Zoom over the WDM comb where are located the three 150-GHz spaced 800-OR channels at the transmitter output in the configuration where the ROADM cascade is inserted in the set-up; (d) Spectrum of the WDM multiplex at the transmitter (red trace) and receiver (blue trace) side after 1500-km; (e) Photograph of the experiment with EVBs and pluggable optics.

At the receiver side, the 800G-OR channel under study (at 1550.12-nm) is extracted from the WDM multiplex by a ~ 175 -GHz flat-top optical band-pass filter and transferred to the coherent receiver (RX) of the 800G-OR OSFP optics from vendor#2 (V#2) or vendor#3 (V#3). These optics are hosted by an EVB from the same vendor and embed a DSP ASIC of the same vendor. The roll-off factor of the RRC filter implemented in these OSFP interfaces is ~ 0.125 [4]. Thus, two DSP chips from different suppliers face each other: V#1 facing V#2 in the first case and V#1 facing V#3 in the second case. Overall, six multi-vendor interoperable configurations are examined, with minor adjustments to the experimental setup described before: (i) V#1 TX $\rightarrow$ V#2 RX, (ii) V#1 TX $\rightarrow$ V#3 RX, (iii) V#2 TX $\rightarrow$ V#1 RX, (iv) V#2 TX $\rightarrow$ V#3 RX, (v) V#3 TX $\rightarrow$ V#1 RX, (vi) V#3 TX $\rightarrow$ V#2 RX. The power consumption in the 800G-OR mode is ~ 25 -Watts for the V#1 QSFP-DD and in the range [27.5-30]-Watts for the V#2 and V#3 OSFP modules.

After optical-to-electrical conversion within the 800G-OR OSFP optic, the V#2 or V#3 EVB extracts the 2x400-GbE client data frames and transmits them to an 800GBASE-2xFR4 QSFP-DD optic coming from the vendor C, which is hosted in the same EVB. The 800GBASE-2xFR4 QSFP-DD interfaces are constituted of two physically separate 400G TX/RX based on four O-band CWDM wavelength lanes at 100G using PAM4 signaling [1] and delivering each ~ 1.3 -dBm. The two ~ 7.3 -dBm 400G signals propagate over two separate 2-km ITU-T G.652 SSMF spools (as depicted in Fig. 1a) before being detected by another 800GBASE-2xFR4 QSFP-DD interface from the vendor D, plugged in a second 800-GbE tester that controls post-FEC errors. Beyond the modulation format and data framing, interoperability is ensured using the KP4 RS(544,514) FEC [1] (Pre-FEC BER threshold $\sim 2.4 \times 10^{-4}$, overhead $\sim 3.03\%$, net coding gain ~ 6.4 -dB at BER $\sim 10^{-12}$), implemented on the host board. The symbol rate and data rate per wavelength, as well as the total power consumption of the 800GBASE-2xFR4 interfaces are 53.125-Gbaud, 106.25-Gbps and ~ 14 -Watts, respectively. A photograph of the experimental set-up is shown in Fig. 1e).

3. Results and discussions

The pre-FEC BER of the 800G-OR signal at 1550.12-nm in the V#1 $\rightarrow$ V#2 configuration is firstly measured in Fig.2 as a function of the channel span input power for various distances. Corresponding OSNRs are simultaneously recorded.

The optimal point for each curve is reported on a BER vs. OSNR graph, which also displays the back-to-back (BtB) performance (Fig. 3). The OSNR value at which the BER vs. OSNR curve reaches the FEC threshold is defined as the required OSNR (ROSNR). According to the OpenROADM MSA [4], the ROSNR at 800G in BtB for Ethernet applications must be lower than 25.1-dB, which is the case here (ROSNR~23.1-dB). More generally, as shown in the table in Fig. 1a), all six configurations under study largely meet this criterion. The optimal span input power for the 800G-OR channel remains stable regardless of the transmission length and is found to be within the [5-6]-dBm range. 800G transmission is achievable up to 15x100-km of SSMF with a received OSNR of ~26.1-dB and an OSNR margin w.r.t. FEC limit of ~0.5-dB (obtained by adding ASE noise on the signal at the receiver). In Fig.4, pre-FEC BERs, OSNRs, and ROSNRs w.r.t. FEC limit are displayed as a function of the transmission length. The difference between the OSNR and ROSNR curves constitutes the OSNR margin w.r.t. FEC limit that is shown in Fig. 5 for each of the six configurations and each distance. The best configuration (V#1→V#2) permits to achieve 1500-km while the less efficient one reaches 1000-km. The transmission performances of bookended configurations are not presented here.

The insertion of a ROADM cascade with two WSSs along the ROADM pass-through path is simulated in the V#1→V#2 configuration by adding a programmable optical filter at the transmitter side for the 1550.12 nm 800G-OR channel. The transfer function of a 150-GHz bandwidth flexgrid WSS is first measured, then cascaded through calculations, and uploaded into the programmable optical filter. Fig. 6 shows the OSNR margins relative to the FEC limit as a function of the number of cascaded WSSs for distances of 1000, 1200, and 1400-km. Up to eight WSSs can be inserted into the 14x100 km transmission line, while up to ten WSSs (i.e., five ROADMs) can be cascaded in the 12x100 km and 10x100 km fiber links, with OSNR margins of ~0.7 dB and ~2.2 dB, respectively.

Finally, long-term performance monitoring was performed on the V#1→V#2 configuration over the 800-ZR, 800G-OR, and 800GBASE-2xFR4 sections, every three minutes over a 24-hour period (Fig. 7). The average pre-FEC BER values measured are $\sim 1.5 \times 10^{-2}$ for the 800-ZR fiber link, $\sim 1.7 \times 10^{-2}$ for the 800G-OR transmission line, and $\sim 6.4 \times 10^{-6}$ and $\sim 3.4 \times 10^{-6}$ for the two 800GBASE-2xFR4 fiber sections. These pre-FEC BERs are well below the FEC thresholds for each section, confirming that the 2x400-GbE client data flows are error-free at the receiver over the 800-GbE tester.

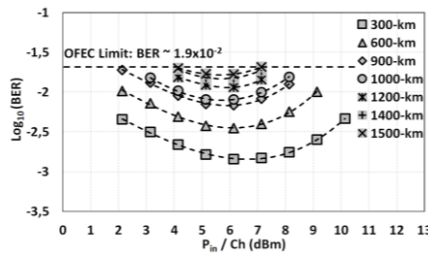


Fig.2: Pre-FEC BER vs. span input power per channel for various distances in the V#1→V#2 configuration for the 800G-OR signal.

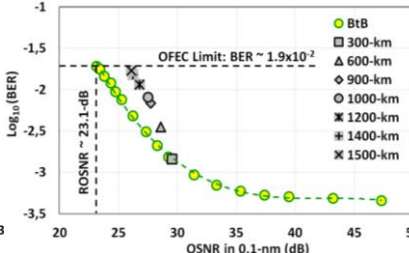


Fig.3: Pre-FEC BER vs. OSNR in BtB, and for various transmission lengths at the optimal span input power determined in Fig. 2.

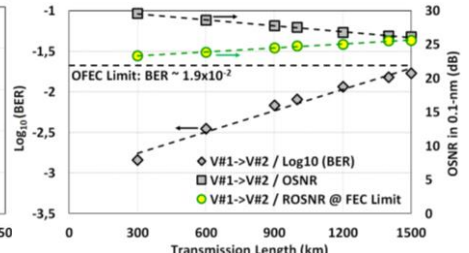


Fig.4: Pre-FEC BER, OSNR and ROSNR w.r.t. FEC limit as a function of fiber length in the V#1→V#2 configuration.

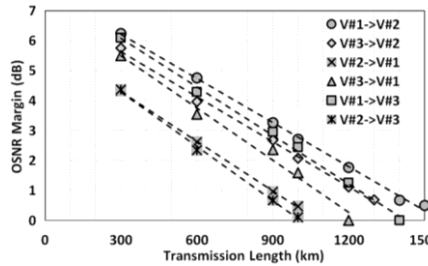


Fig.5: OSNR Margin w.r.t. FEC limit as a function of the transmission distance for the different 800G-OR configurations.

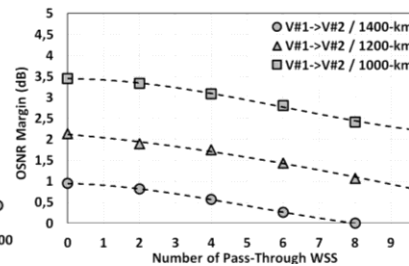


Fig.6: OSNR Margin w.r.t. FEC limit as a function of the cascaded WSSs for the 800G-OR signal in the V#1→V#2 configuration.

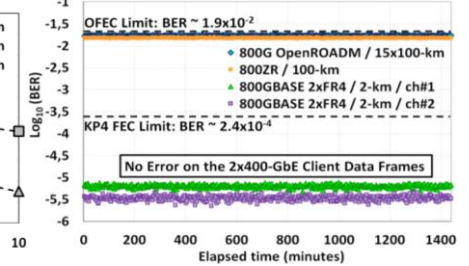


Fig.7: Pre-FEC BER vs. Elapsed Time over 24-hours for the various link sections: 800G-OR (V#1→V#2), 800-ZR, 800GBASE-2xFR4.

4. Conclusion

For the first time, we demonstrated a multi-vendor interoperable 800G optical transmission over 1602 km, fully compliant with the main 800G standards. Specifically, we implemented 800G-OR 131-Gbaud DP-PCS-16QAM coherent optics with the first-ever interoperable PCS used within a multi-vendor environment. Long-term performance measurements confirmed the stability of the proposed solution. This experiment highlights that DCI and metro/regional optical transport networks can benefit from the availability of interoperable 800G coherent optics.

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